

CHLOE & LEO · FALLS CREEK ADVENTURES · BOOK THREE

The Cave of Fair Shares

Deep in the mountain, a treasure that only opens when everything is shared fairly



$\frac{1}{2}$



$\frac{1}{4}$



Day 3 · Fractions · Sharing & Division · Length · Mass · Capacity

The story so far — With Time flowing again, Chloe and Leo follow the Snow Owl's feather to a hidden cave behind a frozen waterfall. Its guardian is a grumbly, fair-minded **Snow Wombat** who will only share his treasure with those who understand *halves*, *quarters*, and *fair shares*. Solve each **DECODE THIS!** to dig deeper!

Behind the Frozen Waterfall

Rocky Valley · the third morning

Behind a curtain of shining ice, Chloe and Leo found a cave. A round, sleepy wombat with a snow-white coat blocked the way, chewing a carrot. "Nobody enters my cave," he grumbled, "unless they can share fairly. Half for you, half for me — no more, no less. Can you even *do* that?"

"Halves are easy!" said Leo. "That's just splitting into two equal parts!"

The Wombat raised one bushy eyebrow. "We'll see about that, small human. We'll see."

— TO BE CONTINUED —

Next: Episode 1 · Halves and Quarters

Halves and Quarters

The mouth of the cave

Previously: A fair-minded Snow Wombat guards a cave and demands everything be shared fairly...

"**A** **half** ($\frac{1}{2}$)," said the Wombat, "is 1 of 2 equal parts. A **quarter** ($\frac{1}{4}$) is 1 of 4 equal parts. Cut my carrot wrong and you get nothing!" A glowing chocolate bar appeared, ready to be shared.

DECODE THIS! — Fair Fractions

Solve all five to satisfy the fussy Wombat.

1. A chocolate bar has 4 equal squares. What is half of it?
2. What is a quarter ($\frac{1}{4}$) of 4 squares?
3. You cut a pizza into 2 equal pieces. Each piece is called a ____ ?
4. Half of 10 marbles is how many?
5. A quarter of 8 lollies is how many?

Solutions: 1) 2 squares 2) 1 square 3) half 4) 5 5) 2

The Wombat sniffed, impressed despite himself. "Hmph. Not bad. But sharing between TWO is easy. Come deeper — where things must be shared between many."

— TO BE CONTINUED —

Next: Episode 2 · The Sharing Stones

The Sharing Stones

A cavern of glowing crystals

Previously: The Wombat is impressed — now the cave opens into a cavern where treasure must be split many ways...

"Dividing," said the Wombat, "is sharing equally into groups. 12 crystals shared between 3 friends — put them out one-by-one until they're gone: 4 each!" Piles of glowing crystals waited to be shared correctly.

DECODE THIS! — Sharing Stones

Share each pile fairly with none left over.

1. Share 12 crystals equally between 3 friends. How many each?
2. Share 10 lollies equally between 2 friends. How many each?
3. Share 15 stickers equally among 5 kids. How many each?
4. 8 apples shared between 4 people — how many each?
5. 20 sweets shared between 4 friends — how many each?

Solutions: 1) 4 2) 5 3) 3 4) 2 5) 5

Every pile split perfectly, glowing bright. "Fairly done," the Wombat admitted. "Now — can you MEASURE fairly too? Follow me to the Hall of Rulers."

— TO BE CONTINUED —

Next: Episode 3 · The Hall of Rulers (Length)

The Hall of Rulers

A long tunnel lined with giant measuring sticks

Previously: Crystals shared, the Wombat leads on to a hall where everything must be measured...

"Length," said the Wombat, "is how long something is. We measure small things in **centimetres (cm)** and big things in **metres (m)**. 100 cm make 1 metre!" Icicles of every size hung from the ceiling, each needing to be measured.

DECODE THIS! — Measuring Length

1. Which would you measure in metres, not centimetres: a pencil, or a ski slope?
2. An icicle is 10 cm long. Another is 5 cm. How much longer is the first?
3. How many centimetres make 1 metre?
4. Leo's ski is 90 cm and Chloe's is 100 cm. Whose is longer?
5. A ribbon is 20 cm. You cut it in half. How long is each piece?

Solutions: 1) a ski slope 2) 5 cm 3) 100 4) Chloe's 5) 10 cm

At the tunnel's end stood a great stone door with ten locks. "The Door of Fair Shares," rumbled the Wombat. "Prove all you've learned, and my deepest treasure is yours to share."

— TO BE CONTINUED —

Next: ⚡ **BIG CHALLENGE** · *The Door of Fair Shares*



BIG CHALLENGE 1

The Door of Fair Shares

A ten-lock stone door · mixes Episodes 1–3

Previously: At the tunnel's end, a ten-lock door tests fractions, sharing, and length together...



THE TEN-LOCK DOOR — solve all ten!

1. What is half of 8?
2. What is a quarter of 12?
3. Share 18 crystals between 3 friends — how many each?
4. Share 16 lollies between 4 friends — how many each?
5. How many cm make 1 metre?
6. An icicle is 14 cm; you break off 6 cm. How long is what's left?
7. Half of 20 sweets is how many?
8. A cake is cut into 4 equal slices. What fraction is 1 slice?
9. Which is longer: 1 metre or 50 cm?
10. Share 21 stickers among 3 kids — how many each?

Door code: 1) 4 2) 3 3) 6 4) 4 5) 100 6) 8 cm 7) 10 8) $\frac{1}{4}$ (a quarter) 9) 1 metre 10) 7

Explorer rank: 8–10 = 🦡 Wombat-Level Fair! · 5–7 = well shared, review a few · under 5 = re-read Episodes 1–3!

The great door split down the middle — *into two equal halves!* — and swung open. Beyond lay a chamber of gleaming scales and jars. "The last tests," said the Wombat. "Weight... and water."

— TO BE CONTINUED —

Next: Episode 4 · The Weighing Chamber

The Weighing Chamber

A room full of golden balance scales

Previously: The Door of Fair Shares opened into a chamber of scales and jars...

"**M**ass," said the Wombat, "is how heavy something is. Light things we weigh in **grams (g)**; heavy things in **kilograms (kg)**. 1000 grams make 1 kilogram!" A golden scale waited to be balanced.

DECODE THIS! — Balancing the Scales

1. Which is heavier: a feather or a snowball?
2. Would you weigh a person in grams or kilograms?
3. One side of a scale has 3 kg. The other has 5 kg. Which side is heavier?
4. A bag holds 2 kg of snow. Another holds 2 kg of feathers. Which is heavier? (Careful!)
5. A snowball weighs 100 g. Two snowballs weigh how much?

Solutions: 1) a snowball 2) kilograms 3) the 5 kg side 4) they weigh the same — 2 kg is 2 kg! 5) 200 g

The scale balanced with a soft *chime*. "One test remains," said the Wombat, leading them to a row of jugs and bottles. "Can you measure water fairly?"

— TO BE CONTINUED —

Next: Episode 5 · The Jugs of the Melt-Water Spring

The Jugs of the Melt-Water Spring

A crystal spring deep in the cave

Previously: The scales balanced; only the water puzzles remain before the treasure...

"Capacity," said the Wombat, "is how much a container can hold. We measure it in **litres (L)**. A big bottle of juice is about 1 litre; a bathtub holds a lot more!"

DECODE THIS! — Measuring Water

1. Which holds more: a teaspoon or a bucket?
2. A jug holds 2 litres. You pour out 1 litre. How much is left?
3. You fill a 1-litre bottle 3 times. How many litres in total?
4. Would you measure a swimming pool in litres or teaspoons?
5. A jug holds 4 litres, shared equally into 2 cups. How many litres per cup?

Solutions: 1) a bucket 2) 1 litre 3) 3 litres 4) litres 5) 2 litres

The Wombat's eyes twinkled. "You have measured, shared, and split everything fairly. You are ready for the Vault of Fair Shares — my greatest test."

— TO BE CONTINUED —

Next:  **GRAND FINALE** · *The Vault of Fair Shares*



GRAND FINALE

The Vault of Fair Shares

The heart of the cave · mixes all of Book Three

Previously: Everything led here — a golden vault with fifteen locks of fractions, sharing, and measuring...



THE FIFTEEN-LOCK VAULT — the final challenge!

Everything from Book Three, easy → hard. Take 30 minutes!

1. What is half of 6?
2. What is a quarter of 8?
3. Share 12 sweets between 4 friends — how many each?
4. Share 20 crystals between 5 friends — how many each?
5. How many cm make 1 metre?
6. How many grams make 1 kilogram?
7. A ribbon is 30 cm; cut in half — how long is each piece?
8. A jug holds 6 litres; pour out 2 — how much is left?
9. Which is heavier: 3 kg or 300 g?
10. A cake in 4 equal slices — what fraction is 2 slices?
11. Half of 14 is how many?
12. Share 24 stickers among 3 kids — how many each?
13. Which holds more: a cup or a bathtub?
14. A quarter of 20 is how many?
15. Two 1-litre bottles and one 1-litre bottle — how many litres in all?

Vault code: 1) 3 2) 2 3) 3 4) 4 5) 100 6) 1000 7) 15 cm 8) 4 litres 9) 3 kg 10) 2 quarters = one half
11) 7 12) 8 13) a bathtub 14) 5 15) 3 litres

Final rank: 13–15 = 🏆 Master of Fair Shares! · 9–12 = 🥈 Brilliant! · 5–8 = 🥉 True explorer · under 5 =
share the cave again, you'll get it!

The vault opened, and inside — not gold, but a feast: enough warm food to share with every hungry animal on the mountain. The Wombat wiped a happy tear. "The real treasure was always *sharing*," he sniffled. "Take this map. Tomorrow you climb the Tower of Numbers." And he handed them a scroll marked with a tall, spiralling tower.

✦ END OF BOOK THREE ✦

Next adventure: Book Four · The Tower of Numbers 🏰



Explorer's Log — Book Three

One ❄️ per riddle cracked. Race or team up!

Episode	Chloe ❄️	Leo ❄️
Ep 1 · Halves & Quarters (/5)		
Ep 2 · Sharing Stones (/5)		
Ep 3 · Hall of Rulers (/5)		
🔒 Door of Fair Shares (/10)		
Ep 4 · Weighing Chamber (/5)		
Ep 5 · The Jugs (/5)		
🏆 Vault Finale (/15)		
TOTAL		